

Qualification Pack



Electronics

QP Code: ELE/Q1502

Version: 2.0

NSQF Level: 4.5

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Qualification Pack

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ELE/Q1502: Electronics

Brief Job Description

This qualification is designed for Electronics Technologists working in Manufacturing and Engineering Technology. It prepares professionals to specialize in areas such as product assembly, circuit design, installation, maintenance, and testing. Key responsibilities include utilizing specialized software for schematic capture, handling a range of tools and equipment, and maintaining professional client interactions with a strong emphasis on confidentiality and ethical practice. The curriculum encompasses essential topics like work organization, electronics applications, prototype hardware design, embedded systems programming, fault diagnosis and repair, as well as assembly and measurement techniques. With a focus on practical skills and lifelong learning, this qualification aligns with international standards and best practices in the rapidly evolving field of Electronics Engineering.

Personal Attributes

The person should be result oriented with good technical and analytical skills, should have Excellent Interpersonal Skills, communication and presentation skills and a good team player. They should have ability to manage projects, prioritizing of work and mentoring the budding engineers.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ELE/N1502: Work organization and management \(Electronics\)](#)
2. [ELE/N1511: Communication and interpersonal skills \(Electronics\)](#)
3. [ELE/N1513: Hardware design and development \(Electronics\)](#)
4. [ELE/N1512: Assembly, modification, and repair \(Electronics\)](#)
5. [ELE/N1506: Fault finding \(Electronics\)](#)
6. [ELE/N1505: Embedded systems programming \(Electronics\)](#)
7. [ELE/N1507: Measurement, documentation, and reporting \(Electronics\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Semiconductor & Components

Qualification Pack

Occupation	System Software Development-S&C
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.0501
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	2.0
Reference code on NQR	QG-4.5-EH-05201-2026-V1-ESSCI
NQR Version	2

Qualification Pack

ELE/N1502: Work organization and management (Electronics)

Description

This NOS unit is about proper training, the correct use of equipment, and adherence to safety protocols in manufacturing settings.

Scope

The scope covers the following :

- Principles and Applications of Safe Working
- Purposes, Uses, Care, and Maintenance of Equipment and Materials
- Environmental and Safety Principles in Good Housekeeping
- Principles of Team Working and Their Applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Plan and organize work tasks based on available time, workflow, and required resources.
- PC2.** Prioritize assignments and resolve challenges using appropriate organizational methods.
- PC3.** Maintain a professional and ethical demeanour in all work-related interactions.
- PC4.** Apply health, safety, and environmental regulations in all workplace activities.
- PC5.** Use tools and equipment correctly while ensuring proper care, storage, and maintenance.
- PC6.** Follow sustainable practices and minimize waste in daily activities.
- PC7.** Reflect on one's work output and identify areas for personal and process improvement.
- PC8.** Engage proactively in professional development to enhance knowledge and skills.
- PC9.** Work responsibly with autonomy while ensuring the wellbeing of oneself and others.
- PC10.** Contribute ideas and actions that support continuous improvement in the organization or project.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of task planning, workflow management, and resource allocation principles.
- KU2.** Understanding of prioritization techniques and problem-solving approaches.
- KU3.** Knowledge of professional ethics and workplace conduct standards.
- KU4.** Understanding of health, safety, and environmental regulations.
- KU5.** Knowledge of proper usage, care, and maintenance of tools and equipment.

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Ability to plan and organize tasks efficiently based on time and resources.
- GS2.** Skill in prioritizing assignments and resolving challenges effectively.
- GS3.** Ability to maintain professional and ethical behaviour in the workplace.
- GS4.** Skill in applying health, safety, and environmental practices consistently.
- GS5.** Ability to use and maintain tools and equipment properly.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Plan and organize work tasks based on available time, workflow, and required resources.	-	-	-	-
PC2. Prioritize assignments and resolve challenges using appropriate organizational methods.	-	-	-	-
PC3. Maintain a professional and ethical demeanour in all work-related interactions.	-	-	-	-
PC4. Apply health, safety, and environmental regulations in all workplace activities.	-	-	-	-
PC5. Use tools and equipment correctly while ensuring proper care, storage, and maintenance.	-	-	-	-
PC6. Follow sustainable practices and minimize waste in daily activities.	-	-	-	-
PC7. Reflect on one's work output and identify areas for personal and process improvement.	-	-	-	-
PC8. Engage proactively in professional development to enhance knowledge and skills.	-	-	-	-
PC9. Work responsibly with autonomy while ensuring the wellbeing of oneself and others.	-	-	-	-
PC10. Contribute ideas and actions that support continuous improvement in the organization or project.	-	-	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1502
NOS Name	Work organization and management (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N1511: Communication and interpersonal skills (Electronics)

Description

This NOS is about to covers the ability to communicate effectively and maintain professional interpersonal relationships to support coordination and teamwork in electronics-related tasks.

Scope

The scope covers the following :

- Covers effective verbal and non-verbal communication, active listening, teamwork, coordination, and maintaining professional relationships in electronics-related work environments.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Maintain professional relationships and communicate effectively with colleagues, supervisors, and external stakeholders.
- PC2.** Use ICT tools for workplace communication, documentation, and external coordination.
- PC3.** Present ideas, proposals, and technical information clearly to internal teams and external groups.
- PC4.** Explain technical concepts in simple terms to clients and non-specialists.
- PC5.** Share knowledge, collaborate in teams, and support collective problem-solving.
- PC6.** Adapt communication style, language, and behaviour based on situational needs while maintaining professionalism.
- PC7.** Challenge inappropriate language or behaviour respectfully, ensuring adherence to workplace ethics.
- PC8.** Maintain accurate records, logs, and documents to ensure traceability and compliance.
- PC9.** Use international symbols, diagrams, and technical languages correctly in communication and documentation.
- PC10.** Comply with ISO and other standardized formats for reports, documentation, and communication.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of professional communication and relationship management in the workplace.
- KU2.** Understanding of ICT tools for communication, documentation, and coordination.
- KU3.** Knowledge of effective presentation techniques for technical and non-technical audiences.
- KU4.** Understanding of methods to simplify technical concepts for non-specialists.
- KU5.** Knowledge of teamwork, collaboration, and knowledge-sharing practices.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Ability to maintain professional relationships and communicate effectively with stakeholders.
- GS2.** Skill in using ICT tools for communication and documentation.
- GS3.** Ability to present ideas and technical information clearly.
- GS4.** Skill in explaining technical concepts in simple and understandable terms.
- GS5.** Ability to collaborate, share knowledge, and support team problem-solving.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Maintain professional relationships and communicate effectively with colleagues, supervisors, and external stakeholders.	-	-	-	-
PC2. Use ICT tools for workplace communication, documentation, and external coordination.	-	1	-	-
PC3. Present ideas, proposals, and technical information clearly to internal teams and external groups.	-	-	-	-
PC4. Explain technical concepts in simple terms to clients and non-specialists.	-	1	-	-
PC5. Share knowledge, collaborate in teams, and support collective problem-solving.	-	-	-	-
PC6. Adapt communication style, language, and behaviour based on situational needs while maintaining professionalism.	-	1	-	-
PC7. Challenge inappropriate language or behaviour respectfully, ensuring adherence to workplace ethics.	-	-	-	-
PC8. Maintain accurate records, logs, and documents to ensure traceability and compliance.	-	1	-	-
PC9. Use international symbols, diagrams, and technical languages correctly in communication and documentation.	-	-	-	-
PC10. Comply with ISO and other standardized formats for reports, documentation, and communication.	-	1	-	-
NOS Total	-	5	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1511
NOS Name	Communication and interpersonal skills (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N1513: Hardware design and development (Electronics)

Description

This NOS unit is about selecting and calculating appropriate component values, designing and modifying circuits to meet specifications, and using simulation software for testing.

Scope

The scope covers the following :

- Calculate and Select Component Values That Are Fit-for-Purpose
- Implement Heatsinking Principles
- Design Modifications to Given Basic Electronics Blocks
- Design Circuits that Meet Specifications and Are Fit for Purpose

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Read and interpret wiring diagrams, schematics, engineering drawings, and technical manuals accurately.
- PC2.** Assist in designing and developing electronic circuits and systems across analog, digital, communication, embedded, and power domains.
- PC3.** Calculate and select appropriate electronic component values based on design specifications.
- PC4.** Use PCB design and schematic capture software to develop circuit schematics and PCB layouts.
- PC5.** Apply 3D PCB visualization tools to verify component placement, mechanical fit, and manufacturability.
- PC6.** Use circuit simulation tools to validate circuit behaviour and ensure designs are fit for purpose.
- PC7.** Conduct research and analysis proportional to the design requirement, applying innovation and creative problem-solving.
- PC8.** Prepare project schedules, cost estimates, bills of materials, and resource requirements based on specifications.
- PC9.** Generate PCB manufacturing data and documentation as per industry standards such as ANSI and IEC.
- PC10.** Evaluate design options considering constraints like time, cost, resources, and functionality, and recommend optimal solutions.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of interpreting wiring diagrams, schematics, and engineering drawings.

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- KU2.** Understanding of electronic circuit design across analog, digital, communication, embedded, and power domains.
- KU3.** Knowledge of component selection and calculation based on design specifications.
- KU4.** Understanding of PCB design and schematic capture tools.
- KU5.** Knowledge of 3D PCB visualization for placement and manufacturability validation.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to read and interpret technical drawings and schematics accurately.
- GS2.** Skill in assisting with the design and development of electronic systems.
- GS3.** Ability to calculate and select appropriate electronic components.
- GS4.** Skill in using PCB design and schematic software effectively.
- GS5.** Ability to verify designs using 3D visualization tools.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	20	-	-
PC1. Read and interpret wiring diagrams, schematics, engineering drawings, and technical manuals accurately.	-	2	-	-
PC2. Assist in designing and developing electronic circuits and systems across analog, digital, communication, embedded, and power domains.	-	2	-	-
PC3. Calculate and select appropriate electronic component values based on design specifications.	-	2	-	-
PC4. Use PCB design and schematic capture software to develop circuit schematics and PCB layouts.	-	2	-	-
PC5. Apply 3D PCB visualization tools to verify component placement, mechanical fit, and manufacturability.	-	2	-	-
PC6. Use circuit simulation tools to validate circuit behaviour and ensure designs are fit for purpose.	-	2	-	-
PC7. Conduct research and analysis proportional to the design requirement, applying innovation and creative problem-solving.	-	2	-	-
PC8. Prepare project schedules, cost estimates, bills of materials, and resource requirements based on specifications.	-	2	-	-
PC9. Generate PCB manufacturing data and documentation as per industry standards such as ANSI and IEC.	-	2	-	-
PC10. Evaluate design options considering constraints like time, cost, resources, and functionality, and recommend optimal solutions.	-	2	-	-
NOS Total	-	20	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1513
NOS Name	Hardware design and development (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	4
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N1512: Assembly, modification, and repair (Electronics)

Description

This NOS is about to covers the ability to assemble, modify, troubleshoot, and repair electronic systems and components in accordance with specifications and standards.

Scope

The scope covers the following :

- Covers assembling, modifying, troubleshooting, and repairing electronic systems and components, including component handling, soldering, testing, fault diagnosis, and ensuring functionality as per specifications.

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and select appropriate tools required for electronic assembly and repair.
- PC2.** Follow IPC 610 and other relevant industry standards during electronic assembly.
- PC3.** Apply IPC 7711/7721 or equivalent standards for rework and repair activities.
- PC4.** Maintain safe working practices in handling tools, components, and equipment.
- PC5.** Implement ESD-safe procedures throughout assembly and repair operations.
- PC6.** Assemble electronic components onto PCBs to create functional circuits.
- PC7.** Identify, assemble, and use electromechanical components and parts.
- PC8.** Form, crimp, route, and assemble cable harnesses as per specifications.
- PC9.** Rework design mistakes to bring assemblies in line with industry standards.
- PC10.** Replace damaged or incorrect components using through-hole and SMD soldering techniques.
- PC11.** Diagnose faulty PCBs to identify defective components or soldering issues.
- PC12.** Repair or replace defective components according to IPC-approved repair techniques.
- PC13.** Conduct soldering, desoldering, cleaning, and inspection as per quality requirements.
- PC14.** Fit or adapt non-original or substitute components/modules to achieve temporary or prototype functionality.
- PC15.** Inspect completed assemblies and verify that workmanship meets industry quality and safety standards.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of tools and equipment used in electronic assembly and repair.
- KU2.** Understanding of IPC 610 standards and industry requirements for assembly quality.
- KU3.** Knowledge of IPC 7711/7721 standards for rework and repair processes.

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KU4. Knowledge of IPC 7711/7721 standards for rework and repair processes.

KU5. Understanding of safe handling practices for tools, components, and equipment.

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. Ability to identify and select appropriate tools for assembly and repair.

GS2. Skill in performing assembly as per IPC and industry standards.

GS3. Ability to carry out rework and repair using standardized techniques.

GS4. Skill in maintaining safe working practices during operations.

GS5. Ability to implement ESD-safe procedures consistently.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	25	-	-
PC1. Identify and select appropriate tools required for electronic assembly and repair.	-	2	-	-
PC2. Follow IPC 610 and other relevant industry standards during electronic assembly.	-	2	-	-
PC3. Apply IPC 7711/7721 or equivalent standards for rework and repair activities.	-	2	-	-
PC4. Maintain safe working practices in handling tools, components, and equipment.	-	2	-	-
PC5. Implement ESD-safe procedures throughout assembly and repair operations.	-	2	-	-
PC6. Assemble electronic components onto PCBs to create functional circuits.	-	2	-	-
PC7. Identify, assemble, and use electromechanical components and parts.	-	2	-	-
PC8. Form, crimp, route, and assemble cable harnesses as per specifications.	-	2	-	-
PC9. Rework design mistakes to bring assemblies in line with industry standards.	-	2	-	-
PC10. Replace damaged or incorrect components using through-hole and SMD soldering techniques.	-	2	-	-
PC11. Diagnose faulty PCBs to identify defective components or soldering issues.	-	1	-	-
PC12. Repair or replace defective components according to IPC-approved repair techniques.	-	1	-	-
PC13. Conduct soldering, desoldering, cleaning, and inspection as per quality requirements.	-	1	-	-
PC14. Fit or adapt non-original or substitute components/modules to achieve temporary or prototype functionality.	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Inspect completed assemblies and verify that workmanship meets industry quality and safety standards.	-	1	-	-
NOS Total	-	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1512
NOS Name	Assembly, modification, and repair (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N1506: Fault finding (Electronics)

Description

This NOS unit is about perform measurements and tests on electronic components using appropriate equipment, isolating faults and repairing components to industry standards.

Scope

The scope covers the following :

- Check the Functionality and Calibration of Test Equipment
- Select the Appropriate Equipment to Perform Measurements
- Take Measurements to Test, Set, Adjust, and Measure Electronic Components, Modules, and Equipment
- Determine Causes of Operating Errors and the Required Action to Repair

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Read and interpret electronic system drawings, schematics, and technical documentation.
- PC2.** Take accurate measurements of electrical quantities (voltage, current, resistance, waveforms) using suitable test instruments.
- PC3.** Select and operate appropriate manual or automated test equipment for diagnostics.
- PC4.** Apply structured fault-finding methods to identify, localize, and isolate faults in electronic circuits and systems.
- PC5.** Determine the root cause of operating errors and carry out necessary corrective actions to restore system functionality.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of interpreting electronic system drawings, schematics, and technical documentation.
- KU2.** Knowledge of manual and automated diagnostic test equipment.
- KU3.** Understanding of structured fault-finding and troubleshooting methodologies.
- KU4.** Knowledge of root cause analysis and corrective action techniques.
- KU5.** Understanding of electrical measurement principles for voltage, current, resistance, and waveforms.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to read and interpret electronic schematics and documentation accurately.

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- GS2.** Skill in measuring electrical parameters using appropriate instruments.
- GS3.** Ability to select and operate diagnostic equipment effectively.
- GS4.** Skill in applying systematic fault-finding techniques to identify and isolate issues.
- GS5.** Ability to analyse root causes and implement corrective actions to restore system functionality.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	5	-	-
PC1. Read and interpret electronic system drawings, schematics, and technical documentation.	-	1	-	-
PC2. Take accurate measurements of electrical quantities (voltage, current, resistance, waveforms) using suitable test instruments.	-	1	-	-
PC3. Select and operate appropriate manual or automated test equipment for diagnostics.	-	1	-	-
PC4. Apply structured fault-finding methods to identify, localize, and isolate faults in electronic circuits and systems.	-	1	-	-
PC5. Determine the root cause of operating errors and carry out necessary corrective actions to restore system functionality.	-	1	-	-
NOS Total	-	5	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1506
NOS Name	Fault finding (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N1505: Embedded systems programming (Electronics)

Description

This NOS unit is about write, compile, and debug C-code for embedded systems, including modifying pre-written code and implementing power management techniques.

Scope

The scope covers the following :

- Locate, Correct, and Re-compile Syntax Errors
- Write, Compile, Upload, Test, and Debug C-Code That Performs to Specification
- Use Common C Functions
- Use Supplied Functions

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Write C-code for embedded systems to execute defined application tasks.
- PC2.** Analyze, review, and update embedded C-code based on requirements or feedback.
- PC3.** Debug code by locating, correcting, and re-compiling syntax and logical errors.
- PC4.** Verify whether the CPU and peripheral devices are responding correctly to program instructions.
- PC5.** Use standard and supplied C library functions appropriately for embedded applications.
- PC6.** Develop new functions/modules to perform specific embedded tasks such as control, data handling, or communication.
- PC7.** Implement microcontroller interfacing routines for common external peripherals.
- PC8.** Apply power-management techniques such as watchdog timers for reliable system operation.
- PC9.** Operate industry-standard Integrated Development Environments (IDEs) for compilation, debugging, and programming.
- PC10.** Perform validation of embedded code through simulation or hardware testing.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of C programming concepts for embedded system applications.
- KU2.** Understanding of code analysis, review, and modification techniques.
- KU3.** Knowledge of debugging methods for identifying and correcting errors.
- KU4.** Understanding of microcontroller architecture, CPU operations, and peripheral interfacing.
- KU5.** Knowledge of standard and embedded C libraries and their usage.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Ability to write efficient C code for embedded applications.
- GS2.** Skill in analysing, reviewing, and updating code based on requirements.
- GS3.** Ability to debug and resolve syntax and logical errors effectively.
- GS4.** Skill in verifying proper functioning of CPU and peripheral devices.
- GS5.** Ability to use standard and custom C library functions appropriately.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	25	-	-
PC1. Write C-code for embedded systems to execute defined application tasks.	-	3	-	-
PC2. Analyze, review, and update embedded C-code based on requirements or feedback.	-	3	-	-
PC3. Debug code by locating, correcting, and re-compiling syntax and logical errors.	-	3	-	-
PC4. Verify whether the CPU and peripheral devices are responding correctly to program instructions.	-	3	-	-
PC5. Use standard and supplied C library functions appropriately for embedded applications.	-	3	-	-
PC6. Develop new functions/modules to perform specific embedded tasks such as control, data handling, or communication.	-	2	-	-
PC7. Implement microcontroller interfacing routines for common external peripherals.	-	2	-	-
PC8. Apply power-management techniques such as watchdog timers for reliable system operation.	-	2	-	-
PC9. Operate industry-standard Integrated Development Environments (IDEs) for compilation, debugging, and programming.	-	2	-	-
PC10. Perform validation of embedded code through simulation or hardware testing.	-	2	-	-
NOS Total	-	25	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1505
NOS Name	Embedded systems programming (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N1507: Measurement, documentation, and reporting (Electronics)

Description

This NOS unit is about assembling the wire electro-mechanical parts, sensors, and surface-mounted devices, ensuring correct sequences and tolerances.

Scope

The scope covers the following :

- Identify and Assemble and Use Electro-Mechanical Parts
- Identify and Assemble Common Sensors
- Assemble Mechanical Parts to Form Working Units
- Wire and Form Cable Harnesses

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare detailed technical drawings and specifications based on engineering instructions.
- PC2.** Draw schematic circuits using schematic capture and PCB layout software.
- PC3.** Produce electronic drawings or graphics for control, instrumentation, sensors, analog/digital systems, and telecom networks using CAD tools.
- PC4.** Prepare professional reports (pdf, doc, xls, etc.) for fault finding, modifications, measurements, and component selection.
- PC5.** Record all calculations performed for component values and electrical quantities with proper units.
- PC6.** Document measurements of circuit parameters such as amplitude, frequency, duty cycle, and cut-off frequency.
- PC7.** Record complete evidence of repairs, including fault description, cause analysis, and corrective actions.
- PC8.** Complete digital reports containing calculations, waveforms, schematics, and required symbols.
- PC9.** Write clear design reports including scheme comparisons, simulation results, and risk analysis findings.
- PC10.** Apply verification standards and use appropriate templates to ensure accuracy and traceability of technical reports.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of preparing technical drawings and specifications as per engineering instructions.
- KU2.** Understanding of schematic capture and PCB layout design tools.
- KU3.** Knowledge of CAD tools for electronic drawings across various systems.

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- KU4.** Understanding of professional reporting formats and documentation practices.
- KU5.** Knowledge of calculation methods for component values and electrical parameters.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to prepare accurate technical drawings and specifications.
- GS2.** Skill in creating schematic circuits using design software.
- GS3.** Ability to produce electronic drawings using CAD tools.
- GS4.** Skill in preparing structured and professional technical reports.
- GS5.** Ability to perform and record calculations with proper units.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	15	-	-
PC1. Prepare detailed technical drawings and specifications based on engineering instructions.	-	2	-	-
PC2. Draw schematic circuits using schematic capture and PCB layout software.	-	2	-	-
PC3. Produce electronic drawings or graphics for control, instrumentation, sensors, analog/digital systems, and telecom networks using CAD tools.	-	2	-	-
PC4. Prepare professional reports (pdf, doc, xls, etc.) for fault finding, modifications, measurements, and component selection.	-	2	-	-
PC5. Record all calculations performed for component values and electrical quantities with proper units.	-	2	-	-
PC6. Document measurements of circuit parameters such as amplitude, frequency, duty cycle, and cut-off frequency.	-	1	-	-
PC7. Record complete evidence of repairs, including fault description, cause analysis, and corrective actions.	-	1	-	-
PC8. Complete digital reports containing calculations, waveforms, schematics, and required symbols.	-	1	-	-
PC9. Write clear design reports including scheme comparisons, simulation results, and risk analysis findings.	-	1	-	-
PC10. Apply verification standards and use appropriate templates to ensure accuracy and traceability of technical reports.	-	1	-	-
NOS Total	-	15	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N1507
NOS Name	Measurement, documentation, and reporting (Electronics)
Sector	Electronics
Sub-Sector	
Occupation	System Software Development-S&C
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training centre (as per assessment criteria below).
4. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training centre based on these criteria.
5. In case of successfully passing only certain number of NOSs, the trainee is eligible to take subsequent assessment on the balance NOS's to pass the Qualification Pack.

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6. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N1502.Work organization and management (Electronics)	-	10	-	-	10	5
ELE/N1511.Communication and interpersonal skills (Electronics)	-	5	-	-	5	5
ELE/N1513.Hardware design and development (Electronics)	-	20	-	-	20	20
ELE/N1512.Assembly, modification, and repair (Electronics)	-	25	-	-	25	25
ELE/N1506.Fault finding (Electronics)	-	5	-	-	5	5
ELE/N1505.Embedded systems programming (Electronics)	-	25	-	-	25	25
ELE/N1507.Measurement, documentation, and reporting (Electronics)	-	15	-	-	15	15
Total	0	105	-	-	105	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.